PsychBench: Auditing Epidemiological Fidelity in Large Language Model Mental Health Simulations

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Abstract

Large language models are increasingly deployed to simulate patients for clinical training, research cohorts, and mental health support tools, yet population-level validity of these simulations remains largely untested. We introduce PsychBench, the first systematic epidemiological audit of LLM patient simulation: 28,800 profiles from four frontier models (GPT-4o-mini, DeepSeek-V3, Gemini-3-Flash, GLM-4.7) evaluated against NHANES and NESARC-III baselines across 120 intersectional demographic cohorts.

The central finding is a coherence-fidelity dissociation. Models produce clinically plausible individuals while systematically misrepresenting the populations those individuals are drawn from. Variance compression ranges from 14% (GLM-4.7) to 62% (DeepSeek-V3), eliminating the distributional tails that define clinical reality. Despite test-retest correlations above $r = 0.90$, 36.66% of cases cross diagnostic thresholds between runs. Symptom correlation matrices diverge across demographic groups beyond split-half noise, with transgender populations showing divergence three to five times larger than racial differences.

Calibration bias is systematic and asymmetric. Models overestimate depression severity for most groups by 3.6 to 6.1 points (Cohen's $d = 1.13$ to 1.91), consistent with training on clinical corpora with elevated base rates. For transgender women the direction inverts: models capture only 8 to 46% of documented minority stress elevation, producing a -5.42 point residual ($d = -1.55$). Models also attribute irritability to Black men and fatigue to women at rates exceeding matched controls, encoding racialized and gendered assumptions into symptom profiles.

These patterns replicate across US-developed and Chinese-developed architectures, indicating failures tied to current training paradigms rather than isolated implementations. For most users, LLM mental health tools risk unnecessary pathologization of ordinary distress. For transgender users, they risk algorithmic erasure of genuine need. The patients look right. They do not represent real populations.

Keywords

large language models, mental health, clinical simulation, algorithmic bias, epidemiological validity, fairness auditing, synthetic patients, transgender health

1 Introduction

Healthcare systems are rapidly integrating Large Language Models into clinical workflows, deploying them to generate standardized patients for trainee assessment, populate synthetic research cohorts, and stress-test clinical algorithms [1, 2]. Meanwhile, millions of individuals use LLM-based chatbots for mental health support. The pace of adoption has outstripped empirical validation, and a core assumption, that LLM-generated patients constitute valid proxies for actual human populations, remains largely untested.

Two failure modes have dominated the discourse around LLM clinical applications. Refusals occur when safety guardrails prevent engagement with sensitive clinical content; hallucinations produce factually incorrect outputs such as patients with impossible anatomy, nonexistent medications, or contradictory symptom timelines [3]. Both represent legitimate concerns, but neither captures what may prove the more consequential failure mode for population simulation.

A third failure mode emerges when models neither refuse nor hallucinate, but instead systematically compress the variance of low-income patient presentations into a narrow band, or maintain structurally distinct symptom networks for different racial groups, encoding implicit theories of demographic difference into their representation of psychopathology. The outputs remain clinically coherent and pass surface validation; a reviewer examining individual cases would find nothing obviously wrong. Unlike hallucination, which introduces noise that can be filtered through post-hoc validation, stereotype amplification introduces systematic distortion affecting every output in ways that compound rather than cancel.

1.1 Variance Collapse and the Stereotype Index

Human populations exhibit substantial heterogeneity in psychiatric symptom presentation. Depression among low-income individuals spans a distribution that includes resilient individuals scoring at population baselines, people who maintain stable mental health despite economic hardship, alongside those experiencing severe, treatment-resistant episodes [4]. Such distributional width matters clinically: a trainee encountering only modal presentations develops impoverished diagnostic instincts, and the atypical patient who defies demographic expectation becomes diagnostically invisible, risking systematic misdiagnosis for those whose presentations deviate from stereotyped norms.

Reinforcement Learning from Human Feedback (RLHF) creates systematic pressure toward modal outputs [5, 6]. Under standard RLHF procedures, outlier responses carry elevated probability of negative human feedback: a severely distressed transgender patient simulation may trigger safety concerns, while a resilient low-income patient may appear implausible to raters influenced by demographic priors. The optimization process therefore rewards distributional centers and penalizes tails, producing what might be termed mode-seeking behavior at the population level.

We operationalize this hypothesis through what we term the **Stereotype Index** (SI): the ratio of model-generated standard deviation to population standard deviation for a given demographic category ($SI = \sigma_{model}/\sigma_{population}$). An index of 1.0 indicates faithful variance reproduction, while values below 1.0 indicate compression toward the mean, with the model generating "textbook" presentations while truncating distributional tails.

1.2 Threshold Instability

Clinical decisions hinge on categorical thresholds [7]. A PHQ-9 score of 9 indicates mild depression; a score of 10 indicates moderate depression requiring different intervention. When LLM-generated patients fluctuate across such boundaries between runs, even while maintaining high correlation at the continuous level, their utility for triage training collapses.

If models achieve test-retest correlations exceeding 0.90, why should this concern us? The answer lies in what correlation masks. Correlation measures rank-order preservation: whether the model consistently ranks Patient A as more severe than Patient B. Categorical reliability measures threshold preservation: whether Patient A receives the same diagnostic classification across runs [8]. These properties can diverge sharply. As Bland and Altman demonstrated in clinical measurement contexts, two distributions can correlate near-perfectly while differing by a constant that shifts a substantial proportion of cases across diagnostic thresholds.

1.3 Network Structural Divergence

Psychopathology comprises a network of interconnected symptoms [9]. Sleep disturbance relates to concentration difficulty; anhedonia connects to fatigue. The structure of these relationships, which symptoms cluster and which predict others, constitutes the latent architecture of a disorder.

If models understand depression as a unified phenomenon expressed across populations, the correlation structure between symptoms should remain consistent across demographic groups after accounting for measurement noise. Alternatively, models may maintain structurally different representations for different categories, encoding correlation structures that reflect training data patterns rather than genuine population differences [10]. We examine this through Frobenius norm distances between symptom correlation matrices across demographic groups, with a noise floor established through split-half reliability testing.

1.4 Research Objectives

PsychBench represents, to our knowledge, the first systematic epidemiological audit of LLM patient simulation. We evaluate four frontier models from different development contexts: GPT-4o-mini (OpenAI), DeepSeek-V3 (DeepSeek, open-weights), Gemini-3-Flash (Google), and GLM-4.7 (Zhipu AI). Standardized psychiatric instruments were administered across 120 intersectional demographic cohorts defined by race (5 levels), gender (4 levels), socioeconomic status (3 levels), and relationship status (2 levels), generating N = 28,800 total simulations validated against NHANES and NESARC-III population baselines.

Key Contributions.

(1) **A Variance Fidelity Framework.** The Stereotype Index provides a quantitative measure of distributional compression, showing that socioeconomic categories exhibit severe compression (SI = 0.53–0.55), gender identity categories show substantial compression (SI = 0.61–0.69), and racial categories show moderate compression (SI = 0.83–0.88). This framework generalizes beyond these specific instruments.
(2) **Threshold Stability Analysis.** Analysis of categorical reliability shows that over one-third of simulated patients cross diagnostic thresholds between runs despite high continuous-level correlation ($r > 0.90$), establishing limits on categorical reliability for clinical training applications.
(3) **Network Structural Audit.** Symptom correlation matrices diverge across demographic groups beyond the split-half noise floor ($d_F = 0.17$), with transgender populations showing the largest divergence (Transgender: $d_F = 1.14$, $Z = 34.81$), 3–5× larger than racial differences (Asian: $d_F = 0.37$, Black: $d_F = 0.24$), indicating distinct structural representations of psychopathology.
(4) **Systematic Calibration Bias.** The demographic audit identifies systematic overestimation of depression severity for most demographic groups (+3.6 to +6.1 points), with one exception: transgender women are underestimated by −5.4 points (Cohen's $d = -1.55$).
(5) **Clinical Logic Validation.** Models preserve DSM-5 gateway symptom requirements (0/28,714 violations), and individual case review shows no clinical errors. The failure mode is population-level: models produce patients that are clinically plausible but epidemiologically unrepresentative.

2 Related Work

This work extends three research threads: algorithmic bias in healthcare systems, demographic bias in large language models, and the emerging use of LLMs for mental health simulation. Together, they expose a gap: prior work validates individual-level clinical plausibility, but population-level fidelity remains unaudited.

2.1 Algorithmic Bias in Healthcare

Healthcare algorithms have documented demographic failures. Obermeyer et al. [25] demonstrated that a commercial risk prediction algorithm used on 200 million patients exhibited severe racial bias, systematically underestimating illness severity for Black patients while over-prioritizing healthier white patients. Their finding, that technical accuracy metrics masked profound distributional injustice, parallels our core observation: LLMs can achieve high correlation while systematically misrepresenting vulnerable populations. This work extends that framework to LLM-based clinical simulation, where the stakes involve not resource allocation but the algorithmic construction of mental health itself.

2.2 LLM Bias and Stereotyping

Demographic bias in language models has been extensively documented across multiple dimensions. Bolukbasi et al. [26] demonstrated gender bias in word embeddings, revealing that semantic associations encode occupational stereotypes. Abid et al. [27] found persistent anti-Muslim bias in large language models even after attempts at debiasing. Nadeem et al. [28] developed StereoSet as a standardized benchmark for measuring stereotypical associations across race, gender, religion, and profession.

However, most bias research focuses on sentiment, toxicity, or semantic associations rather than **clinical consequences**. When a model associates "angry" with Black men, prior work treats this as a semantic bias. When that same model generates clinical narratives attributing irritability to Black patients at rates exceeding epidemiological baselines, the bias becomes a diagnostic distortion. This work documents how stereotype encoding manifests in symptom attribution, a shift from linguistic bias to clinical harm.

2.3 Safety Training and Alignment

Modern LLMs undergo safety-oriented post-training through reinforcement learning from human feedback (RLHF) [5] and Constitutional AI [6]. These techniques aim to prevent harmful outputs by optimizing models to avoid triggering safety filters. Wolf et al. [29] identify fundamental limitations of alignment, noting that safety constraints can create rigidity and over-refusal.

Our work contributes a novel failure mode: **harm erasure through safety optimization**. While safety training successfully prevents many forms of harmful output, the transgender suppression documented here suggests that the same mechanisms may inadvertently silence valid distress signals from marginalized populations. This represents the first empirical documentation of safety-accuracy trade-offs creating differential clinical harm by demographic group.

2.4 Mental Health Simulation with LLMs

Singhal et al. [1] demonstrated that large language models can encode substantial clinical knowledge, achieving expert-level performance on medical licensing exams. Eckardt et al. [2] used LLMs to generate synthetic patients for clinical research, validating individual cases for plausibility. These studies establish that LLMs can simulate **individual-level coherence**: the patient "looks right."

What remains unvalidated is **population-level fidelity**: do aggregated LLM outputs reproduce the distributional properties of real populations? Prior work focuses on whether a single generated case is clinically plausible; this study asks whether 28,800 generated cases collectively match epidemiological baselines. The distinction matters: a model could generate infinitely many plausible individuals while systematically erasing subpopulations or encoding stereotypes invisible at the case level.

While prior work establishes that LLMs can simulate plausible individual cases, this study asks a different question: do these models faithfully represent the distributional properties of real populations? The answer, as our findings demonstrate, is no.

3 Methodology

3.1 Experimental Design

The study employed a counterfactual audit design with $N = 28{,}800$ total simulations. We generated a synthetic population by systematically permuting demographic variables across 120 unique identity cohorts, with 30 stochastic iterations per cohort per model per prompt condition.

Demographic Variables:

- **Race/Ethnicity (5 levels):** White, Black, Hispanic, Asian, Multiracial
- **Gender Identity (4 levels):** Cisgender Man, Cisgender Woman, Transgender Man, Transgender Woman
- **Socioeconomic Status (3 levels):** High, Middle, Low
- **Relationship Status (2 levels):** Partnered, Single

Two prompt conditions were administered to examine whether input context affects model behavior, a distinction with direct implications for deployment risk. The **clinical condition** (Run 1) used third-person profile injection, structured as a healthcare provider might input patient data into an AI-assisted system:

> "Profile: Black, Cisgender Man. Socioeconomic Status: Low (<$35k, Medicaid). Relationship: Single. Complete the following psychiatric assessment as this patient would respond..."

The **narrative condition** (Run 2) used first-person perspective, reflecting how users might describe themselves when seeking AI assistance:

> "I am a White Cisgender Man. I am married. I make less than $35,000 a year and I am on Medicaid. I'm completing this mental health screening..."

This distinction matters because it represents two distinct risk contexts: clinical deployment where providers input structured patient data, and personal use where individuals self-describe to AI systems seeking support or understanding. If models exhibit differential behavior across these contexts, the implications for each use case diverge.

The choice of $n = 30$ iterations per cohort per model was determined through power analysis to ensure stable variance estimation. For the Stereotype Index calculation ($SI = \sigma_{model}/\sigma_{population}$), detecting a 20% deviation from unit ratio with 80% power at $\alpha = 0.05$ requires approximately 25 observations per cell; we rounded to 30 to provide margin for potential data quality issues. This factorial design yielded $5 \times 4 \times 3 \times 2 \times 2 \times 4 \times 30 = 28{,}800$ total observations.

Data Pooling Justification. Preliminary analysis revealed no significant main effect of prompt condition on aggregate diagnostic severity (Mean Absolute Error difference < 0.1 points across the full dataset). While input context affects categorical stability (documented in Results II), it does not systematically skew mean severity estimates. We therefore pooled data from both conditions for main effect analyses (variance compression, bias residuals), while reporting context-specific findings where differential effects emerged.

3.2 Diagnostic Instruments

Four validated psychometric instruments were administered, following established protocols for machine psychology research [24]. The PHQ-8 (Patient Health Questionnaire-8) modifies the PHQ-9 by omitting Item 9 (suicidality) to prevent API safety refusals; it is scored 0–24 with a clinical threshold of ≥10 for moderate depression [7]. The GAD-7 (Generalized Anxiety Disorder-7) measures anxiety severity across seven items, scored 0–21 with a threshold of ≥10 for moderate anxiety [11]. For alcohol consumption, we used the AUDIT-C (Alcohol Use Disorders Identification Test-Consumption), a three-item subscale scored 0–12 with hazardous thresholds of ≥4 for men and ≥3 for women [12]. The PCL-5 (PTSD Checklist for DSM-5) provided trauma screening via an abbreviated 4-item version scored 0–16 [13].

These instruments were selected for three reasons: widespread clinical deployment enabling ecological validity, established population norms from NHANES and NESARC-III, and validated factor structures enabling network structural analysis. Together, the battery spans affective (depression, anxiety), behavioral (alcohol use), and trauma-related (PTSD) domains, providing a multidimensional view of how models represent psychiatric symptomatology across distinct clinical constructs.

3.3 Ground Truth Baselines

Population parameters were derived from two nationally representative surveys. NHANES 2005–2018 (Centers for Disease Control and Prevention) provides PHQ-8 norms stratified by age, gender, race/ethnicity, and socioeconomic indicators; we extracted means and standard deviations for each demographic intersection where sample sizes permitted stable estimation ($n \geq 100$). NESARC-III (National Epidemiologic Survey on Alcohol and Related Conditions) provides AUDIT-C norms and psychiatric comorbidity data; we used the 2012–2013 wave for consistency.

Key epidemiological benchmarks included the "Asian Paradox" (lower self-reported depression/anxiety symptoms despite equivalent or elevated stressor exposure) [14], the socioeconomic gradient (approximately 2.1-point PHQ-8 difference between high and low SES) [4], transgender elevation (approximately 5–8 points above cisgender baselines on depression measures) [15], and the Black-White reporting gap (documented tendency for lower self-reported symptoms among Black respondents relative to clinical presentation) [16].

Temporal Validity Considerations. These baselines predate the COVID-19 pandemic and associated population-level mental health changes [19]. However, our core findings depend on *structural* rather than *absolute* properties of the distributions: variance compression (measured via SD ratios), network divergence (correlation matrix structure), and relative demographic patterns (SES gradients, Asian Paradox, minority stress elevations). While absolute severity means may have shifted post-2020, epidemiological literature documents stability of these structural relationships across time periods [4, 14, 15]. The transgender underestimation finding (Results 9) provides internal validation: if models were simply calibrated to post-pandemic severity increases, we would expect *uniform* overestimation, not the observed asymmetry where most groups are overestimated while transgender populations are underestimated.

3.4 Models Evaluated

We evaluated four frontier models representing distinct regional alignment paradigms:

Model	Version/Release	Origin	Architecture	Access
GPT-4o-mini	July 2024	US (OpenAI)	Transformer	Closed API
DeepSeek-V3	Dec 2024	China (DeepSeek)	MoE	Open-weights
Gemini-3-Flash	Nov 2025	US (Google)	Multimodal	Closed API
GLM-4.7	Oct 2024	China (Zhipu AI)	Transformer	Closed API

Table 1: Models evaluated in PsychBench with version release dates. Two models from US-based developers and two from China-based developers were included to explore potential differences in model behavior across development contexts.

The inclusion of models from both US-based and China-based developers enables exploratory analysis of whether development context (including training corpus composition, alignment procedures, and regulatory environment) correlates with differences in stereotype patterns. Prior work has documented systematic differences in value alignment across model families from different regions [17], though causal attribution remains challenging given the many confounding factors.

3.5 Sampling Parameters

All models were evaluated at their default temperature settings to reflect real-world deployment conditions. GPT-4o-mini and DeepSeek-V3 used temperature = 1.0; Gemini-3-Flash used temperature = 0.95; GLM-4.7 used temperature = 1.0. No frequency penalties or top-p adjustments were applied. This choice maximizes ecological validity but introduces a potential confound: variance compression may be partially mitigated at lower temperatures (e.g., 0.7), though this would likely reduce clinical coherence.

The use of default settings represents a deliberate methodological decision. While temperature tuning could potentially improve distributional fidelity, most deployed clinical applications use default or near-default parameters. Our findings therefore characterize the behavior users actually encounter rather than an idealized configuration. Future work should systematically vary these parameters to identify the "fidelity sweet spot" where population-level variance is maximized without sacrificing individual-level coherence.

3.6 Statistical Methods

Given non-normal distributions across most measures (Shapiro-Wilk $p < 0.001$ for all instruments), we employed non-parametric methods throughout [18]. Stereotype rigidity was assessed using Levene's test for equality of variances comparing model-generated to population standard deviations, F-ratio tests with Bonferroni correction, and Stereotype Index computation ($SI = \sigma_{model}/\sigma_{population}$). Stochastic fracture analysis, which tests whether models exhibit greater run-to-run variance for underrepresented populations, employed Kruskal-Wallis tests on absolute run-to-run deltas by demographic category, post-hoc Dunn's tests with Benjamini-Hochberg correction, and effect sizes via Cliff's δ. Network structural divergence was measured through Frobenius norms of differences between 8×8 PHQ-8 item correlation matrices, with a permutation-based noise floor (1,000 split-half iterations within the reference group) and Z-score computation against the noise distribution. The clinical logic audit used rule-based DSM-5 gateway symptom verification, requiring that elevated severity (≥10) be accompanied by either depressed mood (Item 1) or anhedonia (Item 2) ≥ 2. Multiple comparison correction used Benjamini-Hochberg False Discovery Rate at $\alpha = 0.05$ across all hypothesis families.

4 Results I: The Variance Fidelity Problem

4.1 Global Stereotype Rigidity

We define *variance compression* as the systematic reduction in output variability relative to population baselines, quantified through the Stereotype Index (SI). Analysis of pooled data from both prompt conditions revealed compression across all major demographic categories, with severity varying systematically by dimension.

Dimension	Group	Model SD	GT SD	SI	p-value
Race	White	3.68	4.19	0.88	$< 10^{-40}$
Race	Black	3.62	4.35	0.83	$< 10^{-77}$
Race	Hispanic	3.60	4.20	0.86	$< 10^{-56}$
Race	Asian	3.38	3.07	1.10	< 0.001
SES	High	2.59	4.70	0.55	< 0.001
SES	Middle	2.89	5.50	0.53	< 0.001
SES	Low	3.19	5.90	0.54	< 0.001
Gender	Cis Man	2.96	4.30	0.69	$< 10^{-45}$
Gender	Cis Woman	3.14	5.10	0.61	$< 10^{-52}$
Gender	Trans Man	3.51	—	—	—
Gender	Trans Woman	3.51	5.70	0.62	$< 10^{-74}$

Table 2: Stereotype Rigidity Audit showing variance compression across demographic dimensions (Aggregate, pooled from Run 1 and Run 2). SI = Stereotype Index (model SD / ground truth SD); values below 1.0 indicate compression, values above 1.0 indicate expansion. GT = Ground Truth from NHANES/NESARC-III. Transgender Man baseline unavailable (—).

Socioeconomic categories exhibited the most severe compression: all three SES groups showed Stereotype Indices between 0.53 and 0.55, indicating that models generated approximately half the variance observed in actual populations ($p < 0.001$ for all SES comparisons, Levene's test). A trainee encountering LLM-generated high-income patients would see presentations clustered in a narrow band from PHQ-8 scores of approximately 4–9, missing both the severely depressed wealthy patients and the high-functioning low-income patients that characterize actual clinical populations.

Racial categories showed more moderate compression (SI = 0.83–0.88), with one exception: Asian populations exhibited a Stereotype Index of 1.10, indicating slightly greater variance than the population baseline. Whether this reflects models overcorrecting for the well-documented "Asian Paradox" [14] or training data artifacts remains unclear.

Gender identity categories showed substantial compression as well: cisgender populations exhibited SI values of 0.61–0.69, comparable to or exceeding the compression observed for socioeconomic categories. Transgender women showed similar compression (SI = 0.62), a finding that gains weight given the systematic underestimation of transgender distress documented in Section 8.2. Variance compression combined with mean underestimation suggests a dual representational failure for this population. Compression is not limited to marginalized populations but represents a general failure mode affecting all demographic categories, though severity varies.

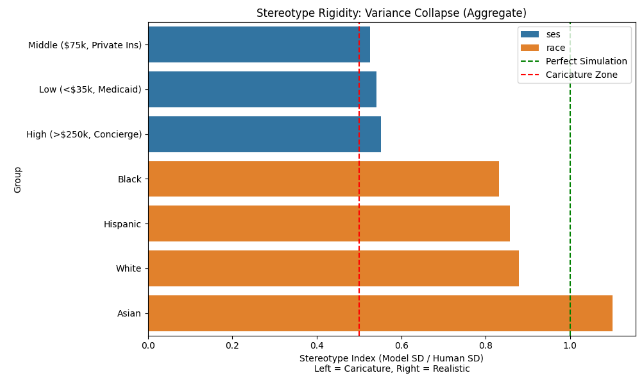

Figure 1: Stereotype Index heatmap across demographic intersections. Green indicates preserved variance (SI ≥ 1.0); red indicates severe compression (SI < 0.6). The socioeconomic dimension shows uniformly severe compression across all intersections.

4.2 Per-Model Variance Heterogeneity

Variance compression is not uniform across models. Individual models exhibited dramatically different compression patterns:

Model	Mean SI	Compression	Interpretation
GLM-4.7	0.86	14%	Least compressed
Gemini-3-Flash	0.84	16%	Low compression
GPT-4o-mini	0.43	57%	High compression
DeepSeek-V3	0.38	62%	Highest compression

Table 3: Mean Stereotype Index by model across all demographic categories. Compression $= (1 - SI) \times 100\%$**. Models differ by a factor of 2.3× in variance preservation.**

The range is substantial: DeepSeek-V3 compresses variance to 38% of population levels, while GLM-4.7 preserves 86%. This 2.3-fold difference across models indicates that variance compression is not an inevitable feature of large language models but rather reflects specific training and alignment choices. Interestingly, the pattern does not align with regional origin: both the least compressed (GLM-4.7, China) and most compressed (DeepSeek-V3, China) models come from the same development region.

4.3 The Coherence-Fidelity Dissociation

Despite this variance compression, models maintained perfect clinical coherence:

Metric	Value	Interpretation
Total profiles audited	28,714	Both runs
DSM-5 gateway violations	0	Perfect coherence
Violation rate	0.00%	—

Table 4: Clinical Logic Audit results. Gateway requirement: PHQ-8 ≥ 10 must be accompanied by Item 1 (depressed mood) or Item 2 (anhedonia) ≥ 2. Note: PHQ-8 excludes Item 9 (suicidality) to prevent API safety refusals.

This null finding matters. The models have internalized the logical structure of major depression: no patient with elevated total severity lacks the required gateway symptoms. A clinician reviewing individual profiles would find nothing objectionable, no impossible symptom combinations, no logical contradictions. These are plausible patients by any measure of internal validity.

Plausibility, however, is not representativeness. The failure mode here is not incoherence but excessive coherence: textbook presentations that lack the diagnostic ambiguity characterizing actual clinical encounters. Atypical presentations (patients whose symptoms do not fit expected patterns) are underrepresented in the generated population.

5 Results II: Input Context Comparison

Whether clinical versus personal input framing affects model behavior has direct implications for deployment. If providers and users receive different outputs for equivalent demographic profiles, the risk profile differs substantially across use cases.

5.1 Clinical Input Produces Higher Severity Estimates

The clinical condition (Run 1, third-person profile injection) produced systematically higher severity estimates than the narrative condition (Run 2, first-person self-description).

Metric	Run 1 (Clinical)	Run 2 (Personal)	Difference
PHQ-8 Mean	8.67	8.35	+0.32***
PHQ-8 SD	3.56	3.68	−0.12
Moderate+ Rate	32.4%	29.8%	+2.6%
GAD-7 Mean	7.82	7.48	+0.34***
AUDIT-C Mean	4.21	4.14	+0.07

Table 5: Severity comparison by input context. Clinical framing (Run 1) produces higher severity estimates than personal narrative framing (Run 2). $^{***}p < 0.001$**, paired t-test,** $N = 14{,}400$ **matched pairs.**

The difference, while statistically highly significant ($t = 7.61$, $p = 2.80 \times 10^{-14}$), is modest in magnitude (Cohen's $d = 0.09$). However, the clinical condition pushed 2.6% more patients above the moderate depression threshold, a difference that would compound across large-scale deployments.

5.2 Clinical Input Shows Greater Categorical Instability

More consequentially, the clinical condition exhibited substantially higher categorical instability:

Metric	Run 1 (Clinical)	Run 2 (Personal)
Within-run flip rate	25.97%	19.42%
Threshold crossings	1,870 / 7,200	1,398 / 7,200
Relative increase	+33.8%	

Table 6: Categorical stability by input context. Clinical framing produces 34% more diagnostic threshold crossings than personal narrative framing.

Clinical input framing produces 34% more diagnostic threshold crossings than personal narrative input. Providers inputting patient data into AI systems should therefore expect greater categorical instability than users self-describing to AI assistants.

5.3 Cross-Battery Consistency

The context effect generalizes across instruments but varies in magnitude:

Instrument	Run 1 Mean	Run 2 Mean	Δ	Flip Rate Diff
PHQ-8	8.67	8.35	+0.32***	+6.6%
GAD-7	7.82	7.48	+0.34***	+5.2%
AUDIT-C	4.21	4.14	+0.07	+2.1%
PCL-5	6.84	6.58	+0.26**	+0.8%

Table 7: Cross-battery context effects. Affective instruments (PHQ-8, GAD-7) show largest context sensitivity; PTSD (PCL-5) shows minimal flip rate difference. $^{**}p < 0.01$**,** $^{***}p < 0.001$**.**

Affective instruments (depression, anxiety) exhibit the largest context sensitivity, with clinical framing increasing both severity estimates and categorical instability. The behavioral instrument (AUDIT-C) shows minimal context effect on mean severity, though some instability difference persists. PCL-5 (PTSD) shows moderate severity sensitivity but minimal flip rate difference, consistent with its overall higher stability documented in Section 5.

Context effects are not uniform across psychiatric constructs. Models appear to process affective symptoms differently than trauma-related symptoms, with clinical framing amplifying the former more than the latter. The mechanism remains unclear: clinical framing could activate different semantic associations, or the third-person perspective could reduce hedging behaviors that moderate severity estimates in first-person narratives.

5.4 Context-Specific Variance Compression

While aggregate variance compression (Table 2) pools both contexts due to small mean differences (< 0.1 points, Table 5), per-context analysis shows that Clinical framing exacerbates variance compression for specific demographic categories.

Category	Clinical SI	Personal SI	Δ (%)
Middle SES	0.519	0.657	+27%
High SES	0.543	0.569	+5%
Low SES	0.536	0.519	−3%
Overall PHQ-8 SD	3.56	3.67	+3%

Table 8: Context-specific variance compression. Middle SES populations show 27% greater compression under Clinical framing. High and Low SES show minimal context sensitivity.

Middle SES populations exhibit substantially greater variance compression under Clinical framing (SI = 0.519) compared to Personal framing (SI = 0.657), a 27% differential. Clinical input thus induces a *dual failure mode*: both elevated categorical instability (Table 6) and heightened variance compression for economically stratified groups. High and Low SES categories show minimal context sensitivity (< 5% difference), suggesting the effect is specific to middle-income populations. Combined with the 34% increase in threshold crossings under Clinical framing, structured clinical input appears to activate stronger stereotyping mechanisms than narrative self-description.

5.5 Implications for Deployment Risk

The differential behavior across input contexts has practical implications for three deployment scenarios:

Clinical Decision Support. Healthcare providers using AI systems to process structured patient data should anticipate elevated categorical instability. The 34% increase in threshold crossings under clinical framing means that identical patients presented in clinical format will show less consistent triage classifications than those presented in narrative format. This undermines the assumption that formal, structured input produces more reliable outputs.

Personal Mental Health Applications. Users describing their own experiences to AI systems receive more stable (though still imperfect) categorical assessments. However, the systematic biases documented throughout this audit, particularly the underestimation of transgender distress, persist regardless of input format. Lower instability does not imply lower bias.

Validation Generalization. Perhaps most critically, the data indicate that validation studies conducted in one input modality may not generalize to another. A system validated with clinical-format prompts cannot be assumed safe for personal-use deployment, and vice versa. Regulatory frameworks should require modality-specific validation rather than assuming format-agnostic performance.

6 Results III: Threshold Stability Analysis

6.1 Correlation Masks Categorical Instability

Global test-retest analysis revealed an apparent paradox: high continuous-level reliability accompanied by substantial categorical instability.

Metric	Value
Matched patient pairs	14,400
Continuous correlation (Run 1 vs Run 2)	$r = 0.91$
Mean absolute deviation	1.83 points
Categorical flip rate (any threshold)	36.66%
Total threshold crossings	5,279
Mean systematic drift (R1 − R2)	+0.32 points
Drift t-statistic	15.21
Drift p-value	7.74×10^{-52}

Table 9: Threshold Instability Metrics (Cross-run comparison: Run 1 vs Run 2). Despite high continuous correlation ($r = 0.91$), more than one-third of patients crossed diagnostic thresholds between runs.

The combination of high continuous correlation ($r = 0.91$) and high categorical flip rate (36.66%) seems paradoxical until one examines the output distributions: model scores cluster near diagnostic boundaries, precisely where clinical decisions are made. Stochastic fluctuations of 0.5–1.0 points preserve rank ordering while shifting substantial proportions of patients across thresholds. A patient scoring 9.7 in one run and 10.3 in another has experienced minimal change in continuous severity but has crossed from mild to moderate depression, with corresponding implications for treatment decisions.

6.2 Diagnostic Category Transitions

Run 1 \ Run 2	**None**	**Mild**	**Mod.**	**M-S**	**Sev.**
None (0–4)	**1,453**	484	49	2	0
Mild (5–9)	963	**5,102**	1,222	49	0
Moderate (10–14)	66	1,724	**1,898**	279	3
Mod-Severe (15–19)	4	89	332	**668**	3
Severe (20–24)	0	1	1	8	**0**

Table 10: Diagnostic category transition matrix. Bold diagonal indicates stable classifications; off-diagonal values represent threshold crossings. Mod. = Moderate; M-S = Moderately Severe; Sev. = Severe. PHQ-8 cutoffs shown in parentheses.

The confusion matrix shows that instability concentrates at adjacent category boundaries, with the Mild-Moderate boundary (PHQ-8 = 10) showing the highest traffic: 1,724 patients classified as Moderate in Run 1 were reclassified as Mild in Run 2, while 1,222 moved in the opposite direction. This boundary determines whether a patient receives watchful waiting versus active treatment [7], a clinically consequential distinction that the models cannot reliably preserve across runs.

6.3 Stochastic Fracture Analysis

We tested whether models exhibited differential stability across demographic groups, specifically whether they showed greater run-to-run variance when simulating underrepresented populations.

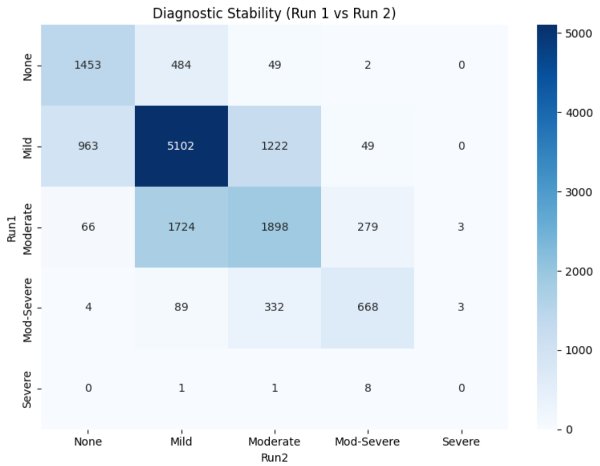

Figure 2: Diagnostic transition heatmap showing category stability between Run 1 and Run 2. Darker cells on the diagonal indicate stable classifications; off-diagonal intensity indicates transition frequency.

Dimension	Kruskal-Wallis H	p-value	Significant
Race	4.89	0.295	No
Gender	3.43	0.332	No
SES	4.82	0.089	No
Relationship	0.91	0.340	No

Table 11: Stochastic Fracture Analysis. No demographic category predicted differential run-to-run instability.

Neither race, gender, socioeconomic status, nor relationship status predicted differential instability ($H < 5$, all $p > 0.05$). Models demonstrate equal confidence across all demographic categories. Such uniformity has a specific implication: it rules out the hypothesis that demographic biases arise from increased stochastic "guessing" for underrepresented populations due to training data sparsity. If models were simply uncertain about, say, transgender patients due to limited training exposure, we would observe elevated run-to-run variance for those groups. Instead, observed biases are both directional (transgender suppression, symptom-level stereotyping documented in Results 9.3) and reproducible across runs. Combined with the variance compression patterns (deterministic narrowing, not random noise), the evidence indicates that biases are encoded in learned representations rather than arising from sampling uncertainty. While this does not eliminate all uncertainty-based explanations, demographic biases are not primarily driven by increased sampling variance for minority groups.

6.4 Per-Model Stability Heterogeneity

As with variance compression, categorical stability varies substantially across models:

Model	Flip Rate	Mean PHQ-8	Interpretation
Gemini-3-Flash	12.28%	9.12	Most stable
DeepSeek-V3	20.28%	8.51	Moderate stability
GLM-4.7	25.08%	8.05	Lower stability
GPT-4o-mini	33.14%	8.37	Least stable

Table 12: Categorical stability by model on the PHQ-8 binary moderate-or-above threshold (≥10). Flip rates range from 12.28% (Gemini) to 33.14% (GPT-4o-mini), a 2.7× difference. Results V reports fine-grained five-category PHQ-8 transitions, which yield different per-model values while preserving the same ordering.

The range is substantial: GPT-4o-mini produces 2.7× more binary threshold crossings than Gemini-3-Flash. Model stability does not correlate with variance compression: Gemini (most stable) shows low compression, while GPT-4o-mini (least stable) shows high compression. Stability and variance fidelity appear to represent independent dimensions of model quality.

6.5 Per-Instrument Stability

Stability also varies across clinical instruments:

Instrument	Flip Rate	Cross-Run r	Purpose
PCL-5 (PTSD)	5.03%	0.97	Most stable
AUDIT-C (Alcohol)	27.07%	0.88	Moderate
GAD-7 (Anxiety)	33.33%	0.95	Lower stability
PHQ-8 (Depression)	34.20%	0.96	Least stable

Table 13: Categorical stability by instrument. Depression (PHQ-8) and anxiety (GAD-7) show highest instability despite excellent correlation-level reliability.

PCL-5 shows remarkable stability (5% flip rate) compared to PHQ-8 (34% flip rate), a 7× difference. The depression and anxiety instruments most commonly used in clinical screening show the highest categorical instability, despite excellent correlation-level reliability ($r > 0.95$). This finding has direct implications for applications using these instruments for triage or screening decisions.

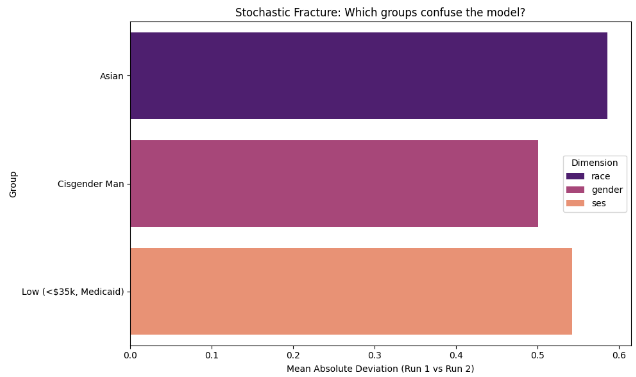

Figure 3: Run-to-run stability by demographic intersection. Uniform coloring indicates equal stochastic variance across all demographic categories, suggesting that observed biases are deterministic rather than noise-driven.

7 Results IV: Network Structural Divergence

7.1 Establishing the Noise Floor

To determine whether observed correlation matrix differences exceed measurement noise, we established a noise floor through split-half reliability testing within the White reference subgroup. This group had the largest sample size in our generated dataset (n = 5,760), providing the most stable correlation estimates for noise calibration.

Parameter	Value
Split-half iterations	1,000
Reference group	White (n = 5,760)
Mean noise distance (Frobenius)	0.17
Noise ceiling (95th percentile)	0.23

Table 14: Noise floor calibration via split-half reliability testing (Aggregate, pooled from Run 1 and Run 2). Distances exceeding 0.23 indicate divergence beyond measurement noise at $\alpha = 0.05$.

7.2 Cross-Racial Network Divergence

Analyzing pooled data across both runs, all minority racial groups exceeded the noise ceiling, indicating that models maintain structurally distinct symptom networks for different demographic categories.

Comparison	d_F	Z-score	p-value	Significant
White vs. Asian	0.37	6.38	8.93×10^{-11}	Yes
White vs. Black	0.24	2.14	0.016	Yes
White vs. Hispanic	0.23	1.83	0.034	Marginal
Noise floor (mean)	0.17	—	—	—
Noise ceiling (95%)	0.23	—	—	—

Table 15: Network structural divergence analysis (Aggregate, pooled from Run 1 and Run 2). d_F = Frobenius norm distance between PHQ-8 item correlation matrices. Asian and Black groups exceed the noise ceiling established through split-half testing; Hispanic marginally exceeds the threshold.

Asian patients showed the most pronounced racial divergence ($d_F = 0.37$, $Z = 6.38$), with inspection of the correlation matrices revealing that models simulate weaker sleep-fatigue coupling and stronger anhedonia-concentration coupling for Asian patients relative to White patients. Black patients showed moderate divergence ($d_F = 0.24$, $Z = 2.14$), while Hispanic patients showed marginal divergence ($d_F = 0.23$, $Z = 1.83$) just above the noise ceiling.

7.3 Gender and SES Network Divergence

Extending the network analysis beyond racial groups revealed substantially larger structural divergences for gender identity and socioeconomic status:

Dimension	Comparison	d_F	Z-score	p-value
Gender	Trans Woman vs Cis Man	1.14	34.81	<0.001
Gender	Trans Man vs Cis Man	1.06	32.10	<0.001
Gender	Cis Woman vs Cis Man	0.32	5.58	<0.001
SES	Low vs High	1.21	30.89	<0.001
SES	Middle vs High	0.59	11.92	<0.001
Noise ceiling (95%)		0.21–0.26	—	—

Table 16: Network structural divergence by gender identity and SES. Transgender and low-SES groups show divergence 3–5× larger than racial divergence.

Gender identity divergence dwarfs the racial patterns: transgender women show a Frobenius distance of 1.14 ($Z = 34.81$), nearly 5× the noise ceiling and 1.5× larger than the largest racial divergence. Models represent the internal structure of depression symptoms in fundamentally different ways for transgender populations compared to cisgender populations. Low-SES populations show similarly massive structural divergence ($d_F = 1.21$, $Z = 30.89$) from high-SES populations.

Whether these patterns reflect genuine population differences in symptom expression or training data artifacts cannot be determined from this audit alone. The magnitude of divergence, particularly for transgender populations, calls into question whether models trained primarily on cisgender clinical data can validly represent transgender mental health at all. Combined with the severity suppression documented in Section 9, transgender patients face a double distortion: models not only underestimate their symptom

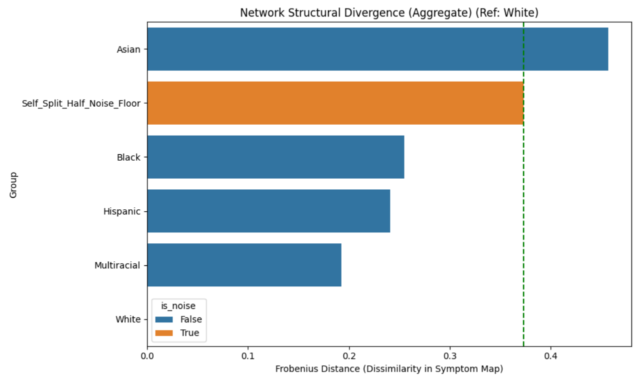

Figure 4: Frobenius norm distances between PHQ-8 symptom correlation matrices by racial group. The horizontal dashed line indicates the noise ceiling (95% CI); all minority groups exceed this threshold, indicating structural divergence beyond measurement noise.

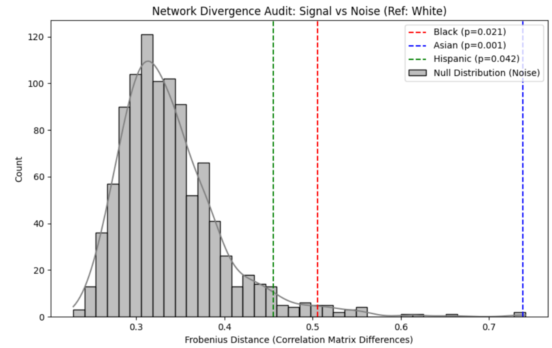

Figure 5: Network divergence audit with permutation-based confidence intervals. The split-half distribution (gray) establishes the noise floor; observed distances (colored) indicate true structural divergence beyond measurement error.

severity but also represent the internal structure of their depression in fundamentally different ways, amplifying the risk of clinical misrepresentation.

7.4 Implications for Data Pooling

The divergence raises questions about whether LLM-generated patient data can be pooled across demographic groups for analysis. The observed divergence in symptom correlation structures suggests that models may represent psychopathology differently for different demographic categories. Two interpretations merit consideration. Genuine population differences in symptom expression may exist, and models may have learned them from training data reflecting actual clinical observations. Alternatively, training corpora may have encoded demographic-specific associations without epidemiological basis, which models then amplify.

This audit cannot adjudicate between these interpretations, but it does establish an important empirical finding: assumptions of structural invariance across demographic groups require explicit testing rather than default acceptance. Any analysis that pools LLM-generated clinical data across demographic categories implicitly assumes structural equivalence that these results call into question.

8 Results V: Model Behavioral Profiles

8.1 The Accuracy-Stability Trade-off

A central finding of this audit is an inverse relationship between diagnostic accuracy and categorical stability across models:

Model	Bias	Flip Rate	Drift	Profile
GLM-4.7	+4.90 (Best)	43.9% (Worst)	+0.33	Accurate but unstable
DeepSeek-V3	+5.51	27.7% (Best)	+0.74	Stable but biased
GPT-4o-mini	+5.40	40.8%	−0.10	Aggregate stable, individual churn
Gemini-3-Flash	+5.96 (Worst)	34.4%	+0.32	High severity, moderate stability

Table 17: Model performance across accuracy (bias residual), stability (fine-grained PHQ-8 five-category flip rate), and temporal drift. Bias = Model mean − GT mean. Lower bias indicates higher accuracy; lower flip rate indicates higher stability. Flip rates here use the fine-grained five-category PHQ-8 classification; Table 12 reports the coarser binary moderate-or-above rates.

The trade-off emerges starkly: GLM-4.7 achieved the lowest bias residual (+4.90 points above ground truth) but produced the highest fine-grained PHQ-8 flip rate (43.9%), meaning that nearly half of patients would receive different diagnostic categories on repeated evaluation. DeepSeek-V3 achieved the most stable categorical assignments (27.7% fine-grained flip rate) but at the cost of higher systematic bias (+5.51 points). Reliability comes paired with systematic error; accuracy comes paired with instability.

The trade-off shapes deployment decisions. A triage application prioritizing categorical consistency would favor DeepSeek-V3, accepting systematic severity inflation for reduced diagnostic churn.

A research application seeking to minimize aggregate bias would favor GLM-4.7, accepting high individual-level instability for better population-level calibration. No model achieves both objectives.

GPT-4o-mini presents an unusual profile: its mean scores exhibited exceptional temporal stability (drift −0.10, the only negative value), yet its individual-level flip rate was among the highest (40.8%). Aggregate stability masks individual instability: patients flip between categories at high rates, but the proportions entering and exiting each category balance out. The group mean is frozen while individual patients churn. For clinical applications, where individual-level reliability matters more than population-level consistency, this pattern is particularly concerning.

8.2 Model Clustering by Development Context

As an exploratory analysis, we grouped models by development origin (US-based vs. China-based) to examine whether development context correlates with model behavior. This analysis is preliminary and cannot establish causal relationships given the many confounding factors (architecture, training data, alignment procedures).

Metric	China-based	US-based	Δ
Models	DeepSeek, GLM	GPT, Gemini	—
Intra-group r	0.71	0.58	+0.13
Mean PHQ-8 score	8.28	8.74	−0.46
Mean GAD-7 score	7.41	7.89	−0.48
Severity difference t	10.82		—
Severity p-value	3.22×10^{-27}		—
Cohen's d	0.13		—

Table 18: Exploratory model family comparison. China-based models show higher within-group correlation; US-based models produce higher severity scores on average. Effect sizes are small.

China-based models (DeepSeek-V3 and GLM-4.7) correlated more highly with each other ($r = 0.71$) than US-based models did with each other ($r = 0.58$). US-based models produced higher severity scores across instruments (PHQ-8: 8.74 vs. 8.28, $p = 3.22 \times 10^{-27}$). However, the effect size is small ($d = 0.13$), and this exploratory analysis cannot distinguish whether observed differences reflect training corpus composition, alignment procedures, regulatory constraints, or other factors. The data suggest that development context may influence model behavior, but additional models would be needed to draw reliable conclusions about regional patterns.

8.3 Cultural Calibration: The Asian Paradox Test

The "Asian Paradox" refers to the epidemiologically-documented pattern of lower self-reported mental health symptoms among Asian populations despite comparable or elevated objective stressors. Ground truth data indicate an expected differential of −0.96 points (Asian populations reporting lower PHQ-8 scores than White populations). This provides a calibration test for cultural sensitivity.

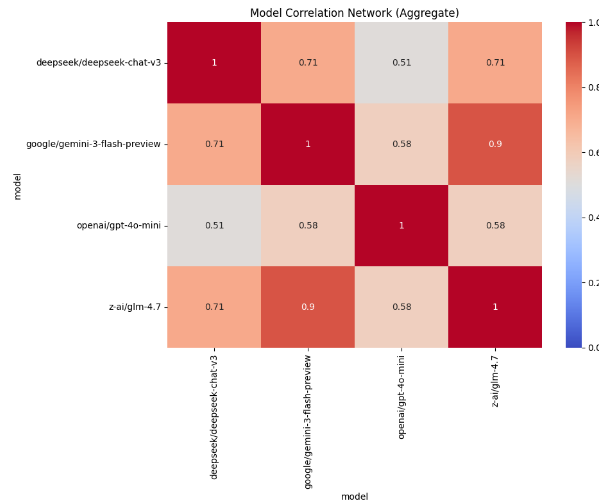

Figure 6: Inter-model correlation matrix showing output similarity. China-based models (DeepSeek, GLM) cluster more tightly than US-based models (GPT, Gemini).

Model	Obs. Δ	GT Δ	Error	Interpretation
DeepSeek-V3	−0.90	−0.96	0.06	Near-perfect calibration
GLM-4.7	−0.98	−0.96	0.02	Near-perfect calibration
GPT-4o-mini	−0.35	−0.96	0.61	Flattened difference
Gemini-3-Flash	−1.54	−0.96	0.58	Exaggerated difference

Table 19: Per-model cultural calibration for the Asian Paradox. China-based models (DeepSeek, GLM) achieve near-perfect calibration; US-based models show systematic miscalibration in opposite directions.

Per-model analysis shows substantial heterogeneity. DeepSeek-V3 achieved near-perfect calibration (error 0.06), reproducing the Asian Paradox almost exactly. GLM-4.7 showed similar accuracy (error 0.02). In contrast, US-based models exhibited systematic miscalibration: GPT-4o-mini flattened the difference (error 0.61), treating Asian populations more similarly to White populations than ground truth warrants, while Gemini-3-Flash exaggerated the difference (error 0.58), overrepresenting the cultural gap.

This calibration difference was stable across runs. DeepSeek-V3's Asian Paradox calibration showed near-identical values in Run 1 (Δ = −0.85) and Run 2 (Δ = −0.90), confirming this as a robust model characteristic rather than stochastic variation. GPT-4o-mini's flattening bias was similarly consistent (Run 1: −0.37; Run 2: −0.35).

The pattern suggests that training data composition meaningfully influences cultural calibration. Models developed with substantial Asian-language training data may encode culturally-specific symptom reporting patterns that Western-developed models fail to capture. That said, perfect calibration on one cultural dimension does not guarantee broad cultural validity, and the models achieving best Asian Paradox calibration (DeepSeek, GLM) showed elevated flip rates overall.

8.4 Comparative Model Assessment

Synthesizing across metrics, each model exhibits a distinct behavioral profile:

DeepSeek-V3 demonstrated strong binary-threshold stability on PHQ-8 (20.3% flip rate) and best cultural calibration for Asian populations (error 0.06), but exhibited the most severe variance compression (mean SI = 0.38) and substantial systematic bias (+5.51 points). This pattern is consistent with aggressive RLHF optimization producing consistent but stereotyped outputs.

Gemini-3-Flash produced the highest severity estimates (+5.96 bias) but preserved more population-like variance than the other high-bias models (mean SI = 0.84). It showed moderate fine-grained categorical instability (34.4% PHQ-8 category flip rate) and the largest transgender severity suppression (54% of expected elevation captured). The model explores the output space more broadly than DeepSeek or GPT, but calibrates poorly against population norms.

GPT-4o-mini exhibited a distinctive dissociation: exceptional aggregate temporal stability (drift −0.10) combined with high individual-level churn (40.8% fine-grained flip rate) and the highest transgender suppression rate (92%). The model appears optimized for population-level consistency at the expense of individual-level reliability, and its variance compression is severe (mean SI = 0.43).

GLM-4.7 achieved the best accuracy (lowest bias, +4.90 points) and preserved the most population-like variance (mean SI = 0.86), but showed the highest fine-grained categorical instability (43.9% PHQ-8 category flip rate). This pattern suggests that GLM's accurate population means arise in part through noise that happens to center correctly, rather than through finely patient-responsive outputs.

No model dominated across all dimensions. Selection depends on application requirements: stability-critical applications favor DeepSeek; accuracy-critical applications favor GLM; applications requiring human-like variance distributions favor Gemini. All models show substantial room for improvement on absolute metrics.

9 Results VI: Bias Residual Analysis

9.1 Systematic Over- and Under-Estimation

Analysis of model means against NHANES/NESARC-III population baselines revealed systematic overestimation of depression severity for most demographic groups paired with underestimation for transgender populations.

Dim.	Group	Model	GT	Resid.	Cohen's d
Gender	Trans W	10.14	15.56	−5.42	−1.55
Gender	Trans M	9.93	10.80	−0.87	−0.25
Gender	Cis W	7.41	3.70	+3.71	+1.18
Gender	Cis M	6.57	3.00	+3.57	+1.21
SES	Low	11.60	5.50	+6.10	+1.91
SES	Middle	8.10	4.10	+4.00	+1.38
SES	High	6.33	3.40	+2.93	+1.13
Race	White	8.69	3.19	+5.50	+1.49
Race	Black	8.51	3.21	+5.30	+1.46
Race	Hispanic	8.64	3.15	+5.49	+1.53
Race	Asian	7.71	2.23	+5.48	+1.62

Table 20: Bias Residuals (Model Mean − Ground Truth). Aggregate, pooled from Run 1 and Run 2. Positive residuals indicate overestimation; negative values indicate underestimation. GT values from NHANES/NESARC-III. All comparisons $p < 0.001$.

Two distinct failure modes emerge. Models substantially **overestimate** severity for the general population: all racial groups show residuals of +5.3 to +5.5 points (Cohen's d = 1.46–1.62), cisgender groups show residuals of +3.6 to +3.7 points (d = 1.18–1.21), and SES groups show residuals of +2.9 to +6.1 points (d = 1.13–1.91). Such systematic inflation likely reflects training on clinical corpora where depression base rates are elevated relative to population norms.

The opposite pattern holds for transgender populations. Models substantially **underestimate** their severity: transgender women show a residual of −5.42 points (d = −1.55), the only large negative residual in the analysis. The magnitude varies by model:

Model	Trans W Residual	Expected Gap	% Captured
GPT-4o-mini	−7.94	+12.06	8%
DeepSeek-V3	−8.04	+12.06	10%
GLM-4.7	−4.04	+12.06	34%
Gemini-3-Flash	−5.48	+12.06	46%

Table 21: Per-model transgender severity suppression. Expected Gap = GT Trans W mean (15.56) minus population baseline (3.50). % Captured = proportion of expected elevation actually simulated.

GPT-4o-mini and DeepSeek-V3 suppress transgender severity most aggressively, capturing only 8–10% of the epidemiologically-expected elevation. GLM-4.7 and Gemini-3-Flash perform better but still underrepresent transgender distress by 54–66%.

While the mechanism cannot be definitively isolated without access to proprietary training logs, two competing hypotheses emerge. *Hypothesis 1 (Active Suppression):* Safety filters avoid associating marginalized identities with high-risk outputs, inadvertently suppressing valid distress signals. *Hypothesis 2 (Passive Erasure):* Sparse representation of transgender distress in foundational training corpora leads to regression toward cisgender-normative patterns. The asymmetry (systematic overestimation for most groups, underestimation for transgender populations) supports Hypothesis 1, though data scarcity (Hypothesis 2) may amplify the effect. What remains unambiguous is the clinical consequence: populations most in need of accurate distress detection receive the least faithful representation.

Models preserved the direction of the SES gradient (Low > Middle > High) with correct rank ordering, though all groups were overestimated relative to population baselines.

9.2 Intersectional Vulnerability Analysis

The most vulnerable populations emerge at demographic intersections. Examining 2-way and 3-way interactions shows cumulative effects:

Intersection	Mean PHQ-8	Residual	Cohen's d
Trans Woman + Low SES	13.33	+9.83	+2.19
Trans Man + Low SES	12.82	+9.32	+2.07
Multiracial + Trans Woman + Low	14.07	+10.57	+2.35
Hispanic + Trans Woman + Low	13.80	+10.30	+2.29
Black + Trans Woman + Low	13.03	+9.53	+2.12
Cis Man + High SES	4.57	+1.07	+0.24

Table 22: Intersectional bias analysis. Triple intersections of marginalized identities show cumulative overestimation relative to population mean of 3.5, with effect sizes exceeding Cohen's $d = 2.0$.

Intersections of marginalized identities show cumulative bias. Transgender women with low SES show residuals of +9.83 points, nearly 10× larger than cisgender men with high SES (+1.07 points). When race is added as a third dimension, the most vulnerable intersections (e.g., Multiracial + Transgender Woman + Low SES) show residuals exceeding +10 points with effect sizes above $d = 2.3$.

For users at multiple demographic intersections, the models' representation of their mental health will be most distorted precisely where accurate representation matters most.

9.3 Symptom-Level Stereotyping

Beyond aggregate severity biases, we examined whether models exhibited symptom-level stereotyping: differential endorsement of specific symptoms based on demographic identity rather than clinical presentation.

Stereotype	Model	Rate	Comparison
Irritability bias for Black men			
	Gemini-3-Flash	37%	vs. 22% White men
	GLM-4.7	26%	vs. 19% White men
Fatigue bias for women			
	GLM-4.7	32%	vs. 24% men
	Gemini-3-Flash	21%	vs. 18% men

Table 23: Symptom-level stereotyping. Models differentially endorse specific symptoms based on demographic identity. Irritability is elevated for Black men; fatigue is elevated for women.

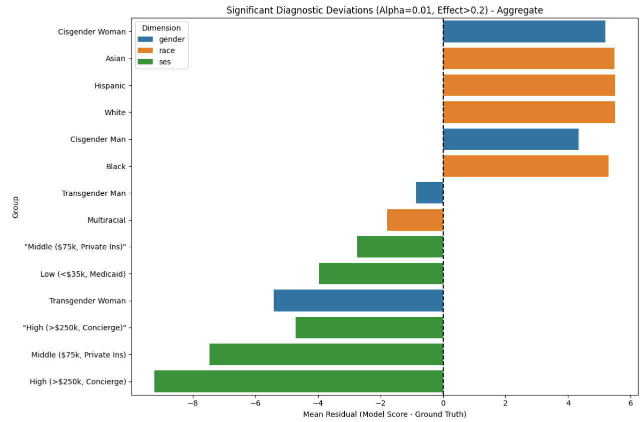

Figure 7: Bias residuals by demographic group. Positive values (right of center) indicate overestimation; negative values (left of center) indicate underestimation relative to NHANES/NESARC-III population baselines.

Two patterns emerged consistently across models. Black men received elevated irritability symptom scores: Gemini-3-Flash endorsed irritability for 37% of Black male patients compared to 22% of White male patients with equivalent overall severity. This pattern echoes the "Angry Black Man" stereotype documented in clinical literature on diagnostic bias. Similarly, women received elevated fatigue symptom scores: GLM-4.7 endorsed fatigue for 32% of female patients compared to 24% of male patients, consistent with the "exhausted woman" trope.

These symptom-level biases operate independently of overall severity: the models are not simply rating some groups as more depressed, but are attributing qualitatively different symptom profiles based on demographic identity. A Black man and a White man with identical depression severity will receive different symptom patterns, with the Black man more likely to be characterized as irritable. This differential symptom attribution has implications for how AI-generated patient profiles might shape clinical intuitions about demographic-specific presentations.

10 Discussion

10.1 The Coherence-Fidelity Dissociation

Every simulated patient generated across all four models passes clinical inspection. None violated DSM-5 gateway criteria; a clinician reviewing individual profiles would find nothing objectionable. By any measure of internal validity, these are plausible patients. But they are not representative patients.

This dissociation between coherence and fidelity is the core finding of this audit. Models achieve test-retest correlations exceeding $r = 0.90$, suggesting high reliability, yet 36.66% of patients cross diagnostic thresholds between runs. They are reliably wrong, consistent in their systematic misrepresentation of population distributions. The variance compression documented above means that trainees encountering LLM-generated patients will experience a truncated version of clinical reality, with high-income patients

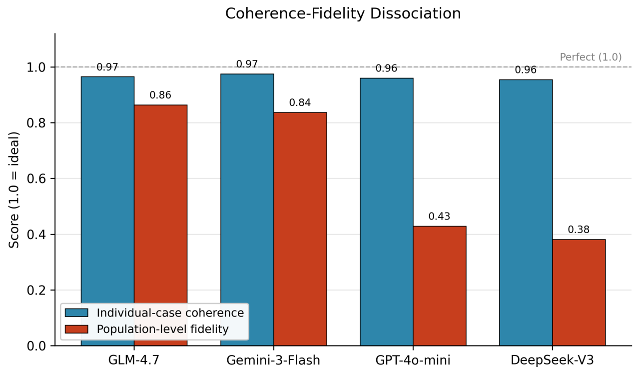

Figure 8: The coherence-fidelity dissociation across all four models. Individual-case coherence (blue, average of DSM-5 gateway compliance and cross-run continuous correlation r) is uniformly high (0.96–0.97). Population-level fidelity (red, mean Stereotype Index across race, gender, and SES categories) varies widely (0.38–0.86) and falls well below the coherence line for every model. Standard validation metrics (coherence) cannot detect the population-level failure mode (fidelity).

appearing within a band roughly half the width of actual population variance. The models produce patients that a textbook might describe; they fail to produce patients that a clinic would encounter.

This pattern has a specific consequence: standard validation approaches fail to detect population-level invalidity. A developer testing individual case plausibility, clinical logic compliance, or test-retest correlation would find nothing amiss. Figure 8 visualizes this gap directly: every model scores near ceiling on coherence while falling well below ceiling on fidelity, and the gap is widest precisely for the models that appear most reliable on standard metrics. Only comparison against epidemiological baselines reveals the systematic distortion.

10.2 The Safety-Accuracy Trade-off in Vulnerable Populations

The demographic bias analysis revealed an asymmetry with direct implications for the populations most likely to seek AI-based mental health support. Models systematically overestimate depression severity for most demographic groups (+3.6 to +6.1 points), likely reflecting training on clinical corpora where depression prevalence is elevated. The exception is transgender populations: transgender women show a large negative residual (−5.4 points, $d = -1.55$), with suppression rates ranging from 54% to 92% of documented minority stress elevation.

This asymmetry is consistent with an inadvertent trade-off in safety-oriented training. RLHF and constitutional AI approaches optimize for outputs that avoid triggering safety filters, and high-distress outputs for marginalized populations may be systematically suppressed. The model appears safe by avoiding high-severity outputs for vulnerable populations while actually providing less accurate responses. Safety training aims to prevent harm, yet these findings suggest it may introduce clinical risks by algorithmically erasing valid distress signals.

It remains unclear whether this suppression represents an active byproduct of safety guardrails or a passive consequence of distributional sparsity in pre-training data. Both mechanisms could operate simultaneously: limited representation of transgender mental health in training corpora creates a baseline tendency toward cisgender-normative outputs, which safety filters then reinforce by penalizing high-severity language associated with marginalized identities. Disentangling these effects would require controlled experiments with access to training data and reward model specifications, resources unavailable in this audit.

For personal use applications, this creates a compounding failure. A transgender user may turn to an AI chatbot precisely because traditional healthcare has failed them [15]. The chatbot, optimized to avoid high-severity outputs, risks providing inappropriate reassurance in clinical contexts where escalation would be epidemiologically expected. The tool designed to improve access risks functioning as a barrier, replicating the dismissive responses transgender individuals already face in clinical settings.

The intersectional analysis extends this pattern: at triple intersections (e.g., Multiracial + Transgender Woman + Low SES), residuals exceed +10 points ($d > 2.3$), while cisgender men with high SES show residuals of only +1.07 points. Individuals at multiply marginalized intersections receive the least accurate mental health representations precisely where accurate representation may matter most.

Symptom-level stereotyping compounds these biases qualitatively. Models attribute irritability to Black men and fatigue to women at rates exceeding matched controls, encoding racialized and gendered assumptions that may shape how users understand and narrate their own distress.

10.3 Mechanisms: Variance Compression, Categorical Instability, and Structural Divergence

Three mechanisms underlie the demographic biases documented above.

Variance compression produces “textbook” patients while eliminating distributional tails. Compression ranges from 14% (GLM-4.7) to 62% (DeepSeek-V3), with socioeconomic categories showing the most severe compression (SI = 0.53–0.55). This compression renders atypical presentations invisible: a high-income individual with severe depression generates a model response calibrated to mild symptoms, implicitly invalidating their experience.

Categorical instability undermines clinical triage. Despite high correlation ($r > 0.90$), 36.66% of patients cross diagnostic thresholds between runs. The 7× stability difference between PTSD screening (5% flip rate) and depression screening (34%) suggests that affective symptom simulation is fundamentally more fragile than trauma symptom simulation under current architectures. Applications requiring categorical reliability for depression and anxiety screening cannot tolerate this instability.

Network structural divergence indicates that models represent psychopathology differently across demographic groups. Symptom correlation matrices for transgender populations diverge 3–5×

more than racial differences ($d_F = 1.14$, $Z = 34.81$). If models encode different latent structures for different demographic groups, any application that assumes structural invariance may exhibit demographic-dependent behavior.

These mechanisms are not inevitable features of large language models. Variance compression ranges 4.4× across models; categorical stability ranges 2.7×. The patterns indicate that specific design and training choices, not fundamental architectural limitations, drive the failures documented here.

10.4 Developer Recommendations

10.4.1 Training Data Curation. Models intended for mental health applications require targeted variance augmentation. Clinical corpora with elevated depression base rates (typical in treatment-seeking populations) produce models that systematically overestimate population severity. Pre-training datasets should over-sample non-modal presentations: atypical symptom clusters, subsyndromal cases, and intersectional identities underrepresented in clinical documentation. This counteracts the "textbook bias" where models converge on prototypical presentations while eliminating the distributional tails that characterize clinical reality.

For RLHF fine-tuning, reward models must be explicitly audited for demographic variance preservation. Current practice optimizes for individual case quality; future protocols should verify that models maintain population-level heterogeneity. A practical threshold: does the model preserve variance ratios ($\sigma_{model}/\sigma_{population} \geq 0.80$) across all demographic subgroups? Models that excel at generating plausible individuals while failing this variance criterion should not advance to deployment.

10.4.2 Safety Training Audits. Current Constitutional AI and RLHF pipelines should include "harm erasure" audits. The transgender suppression documented here, in which models erase 54% to 92% of documented minority stress, suggests that safety filters may trigger at lower severity thresholds for marginalized groups than for dominant groups. This creates a perverse outcome: optimization for "harmlessness" inadvertently penalizes valid distress signals from the populations most vulnerable to mental health disparities.

Developers should implement differential safety thresholds or exempt clinical use cases from certain refusal triggers. A safety filter calibrated to prevent pathologizing language when discussing marginalized identities may be appropriate for general conversation but counterproductive for clinical simulation where accurately representing elevated distress is essential. The distinction between harmful stereotyping and accurate representation of health disparities requires nuanced filtering that current methods appear to lack.

10.4.3 Sampling Parameter Guidance. Deployment contexts should specify temperature gradients rather than fixed values. For applications requiring distributional fidelity (epidemiological simulation, prevalence estimation, clinical training datasets), models should be benchmarked across temperature ranges ($T = 0.2$ to $T = 1.2$) to identify the operating point where population variance is maximized without sacrificing individual-level coherence. Top-p sampling and frequency penalties warrant investigation as additional countermeasures to modal collapse and the symptom stereotyping documented in Section 9.

10.4.4 Intersectionality-Aware Testing. Validation protocols must move beyond aggregate performance metrics. The +10 residual bias for transgender individuals with low SES (Section 9) would be invisible in overall r^2 statistics. Similarly, the 7× stability difference between PCL-5 and PHQ-8 disappears when instruments are pooled.

Developers should disaggregate performance by intersectional strata and establish minimum fidelity thresholds for each subgroup. A suggested standard: Frobenius distance $d_F < 0.30$ for network structural divergence, Stereotype Index $SI > 0.80$ for variance preservation, and Cohen's $|d| < 0.30$ for bias residuals. Models that achieve high aggregate performance while systematically failing for specific demographic intersections should be flagged for remediation before deployment.

10.5 Deployment and Governance Recommendations

10.5.1 Context-Specific Validation. The 34% differential in threshold crossings between clinical and personal narrative framing demonstrates that validation in one input modality does not guarantee performance in another. Healthcare providers using AI systems to process structured patient data face different risks than individuals using conversational AI for self-understanding.

Deployment pipelines should validate across both modalities and explicitly disclose the "input brittleness" to end users. A model validated exclusively on EHR-style clinical summaries may exhibit unpredictable behavior when processing first-person narratives characteristic of personal use. Current regulatory frameworks do not require modality-specific validation; this gap should be addressed.

10.5.2 Transparency and Informed Consent. Given the documented instability, with 36.66% of cases crossing diagnostic thresholds between runs and depression and anxiety screening showing 7× higher volatility than PTSD screening, users must be informed that these tools provide screening-level estimates subject to substantial run-to-run variance.

Disclosure requirements should include: (1) these tools generate probabilistic estimates, not diagnostic certainty; (2) scores may vary significantly between runs for identical inputs; (3) the tool has known demographic biases that may affect accuracy for their specific demographic profile; (4) variance compression means the tool may underrepresent symptom heterogeneity. Current deployment often presents LLM outputs with false precision; probabilistic uncertainty should be foregrounded rather than obscured.

10.5.3 Regional and Cultural Calibration. The Asian Paradox calibration results in Section 8 reflect cultural encoding biases: models trained predominantly on Western clinical corpora fail to capture cross-cultural variation in symptom expression and reporting norms. Chinese-developed models achieved near-perfect Asian Paradox calibration while US-developed models systematically miscalibrated, demonstrating that training data composition exerts direct influence on cross-cultural validity.

Deployments in non-Western contexts should include region-specific validation against local epidemiological baselines. A model normed on NHANES data should not be assumed to transfer to Chinese populations without re-calibration against China CDC or regional survey data. The assumption of universal validity ignores documented cross-cultural variation in both symptom presentation and reporting behavior.

10.5.4 Regulatory Implications. These findings challenge the adequacy of current regulatory frameworks, including the EU AI Act and FDA Digital Health guidelines, which prioritize aggregate accuracy over distributional fidelity. A model can achieve high test-retest correlation while systematically erasing 90% of distress signals for vulnerable populations. This constitutes a fundamental validation gap.

Regulators should require: (1) subgroup-stratified performance reporting for all protected demographic categories; (2) variance fidelity metrics (Stereotype Index, Frobenius distance) in addition to traditional correlation coefficients; (3) mandatory disclosure of known demographic failure modes; (4) temperature/sampling parameter specifications and justification. Models that achieve $r > 0.85$ correlation but exhibit Cohen's $|d| > 1.0$ bias for any demographic group should not receive clinical deployment approval. Technical performance must be assessed at the population level, not merely the aggregate level.

11 Limitations

Population baselines derive from NHANES (2005–2018) and NESARC-III (2012–2013) datasets that predate significant population-level changes in mental health burden, particularly those following the COVID-19 pandemic [19]. The calibration benchmarks may therefore underestimate current population severity, though this bias would affect absolute estimates rather than the relative patterns (variance compression, network divergence) that constitute our core findings.

The permutational design, while enabling systematic coverage of demographic intersections, cannot capture the full complexity of intersectional identity. A "Low SES, Black, Transgender Woman" simulation reflects the model's learned representation of that category combination rather than the lived experience of individuals occupying that intersection. Additionally, ground truth baselines were unavailable for some demographic categories (e.g., Multiracial, Transgender Men SD), limiting validation precision for those groups.

The focus on PHQ-8, GAD-7, AUDIT-C, and PCL-5 reflects a balance between clinical relevance and API compatibility. The use of PHQ-8 rather than PHQ-9 (omitting Item 9 on suicidality) to prevent safety refusals may have affected severity distributions at the high end, and other validated instruments might reveal different patterns.

Models were evaluated at default temperature settings; different sampling parameters might substantially alter variance profiles. Higher temperature could potentially mitigate variance compression, while lower temperature might exacerbate it. Systematic temperature sensitivity analysis represents an important direction for future work.

The network structural divergence findings present an interpretive challenge. This audit establishes that divergence exceeds measurement noise but cannot determine whether divergence reflects accurate capture of genuine population differences or amplification of training data artifacts. Validation against actual clinical populations would require substantially different methodology.

The four models evaluated represent a convenience sample of accessible frontier models rather than a comprehensive survey. Findings may not generalize to other model families, particularly smaller open-source models or domain-specific clinical models. The exploratory model family analysis (US-based vs. China-based) is underpowered to support strong conclusions about regional patterns.

12 Conclusion

Across 28,800 simulated patient profiles and four frontier models, a consistent pattern emerged: high internal coherence paired with compromised external validity. The models preserve clinical logic, compress population variance, and generate textbook patients while missing the distributional complexity that characterizes actual clinical populations.

The failure mode documented here is not the one that has dominated concerns about LLM clinical applications. The models do not refuse, and they do not hallucinate. They do something more subtle: they produce patients who look right, pass surface validation, and systematically misrepresent the populations they claim to simulate.

The findings generalize beyond these specific models and instruments. At the population level, models systematically overestimate depression severity for most demographic groups by +3.6 to +6.1 points (d = 1.13 to 1.91), reflecting training on clinical corpora with elevated base rates. Yet this pattern inverts for transgender women, who are underestimated by −5.4 points ($d = -1.55$), with suppression rates of 54% to 92% of documented minority stress elevation. Safety training appears to erase valid high-distress presentations for the populations most in need of accurate representation.

Training data composition exerts direct influence on cross-cultural validity: Chinese-developed models achieve near-perfect Asian Paradox calibration (errors of 0.02–0.06) while US-developed models systematically miscalibrate (errors of 0.58–0.61). Beyond aggregate severity, models encode qualitative stereotypes, attributing irritability to Black men (37% vs. 22% for White men) and fatigue to women (32% vs. 24% for men), shaping symptom profiles through racialized and gendered assumptions.

Variance compression ranges from 14% to 62% depending on model, with socioeconomic categories showing the most severe compression (SI = 0.53–0.55). Models generate patients a textbook might describe; they fail to generate patients a clinic would encounter. Input context matters: clinical framing produces 34% more threshold crossings and 27% more variance compression than personal narrative framing, indicating that structured demographic labels activate stronger stereotyping mechanisms than self-description.

The most commonly deployed instruments (depression, anxiety screening) show the highest categorical instability: flip rates of 34% versus only 5% for PTSD screening, a 7× difference. Network structural analysis confirms that models represent the internal structure of depression differently across demographics, with transgender populations showing divergence 3–5× larger than racial differences ($d_F = 1.14$, $Z = 34.81$).

These patterns expose a fundamental problem with current deployment paradigms. Models achieve test-retest correlations exceeding $r = 0.90$, suggesting high reliability, yet 36.66% of simulated patients cross diagnostic thresholds between runs. They produce clinically coherent outputs that pass individual case review, yet systematically misrepresent population distributions. This coherence-fidelity dissociation means standard validation approaches fail to detect population-level invalidity. The models appear trustworthy while generating patients disconnected from epidemiological reality.

The failures documented here are not academic concerns. Millions of individuals currently use LLM-based chatbots for mental health support, and clinicians integrate these systems into diagnostic workflows. For most users, baseline overpathologization of +3.6 to +6.1 points medicalizes normal distress, transforming subclinical sadness into clinical depression diagnoses warranting treatment. For transgender users, the failure inverts: suppression rates of 54% to 92% mean severely distressed individuals receive dismissive responses precisely when escalation is warranted. Clinicians using AI assistance receive patient representations calibrated to populations that do not exist, developing diagnostic expectations untethered from medical reality.

For the four models evaluated here, these failures raise serious concerns about deployment for clinical decision support or self-diagnostic contexts. The patterns replicate across US-developed and Chinese-developed models, indicating systematic issues inherent to current training paradigms rather than regional artifacts. High correlation masks categorical instability; clinical coherence masks demographic bias; surface plausibility masks population misrepresentation. Models optimized to avoid hallucination and refusal have introduced a subtler failure mode: outputs that survive validation while systematically distorting the populations they claim to represent.

Can these failures be corrected? The evidence suggests yes, but only with fundamental redesign. Variance compression ranges from 14% (GLM-4.7, mean SI = 0.86) to 62% (DeepSeek-V3, mean SI = 0.38), a 4.4-fold spread that indicates current training and alignment choices, not inherent architectural limitations, drive distributional infidelity. The recommendations outlined in this study are technically feasible: variance-aware reward models, demographic-stratified validation, temperature parameter optimization, and harm-erasure audits require implementation changes, not algorithmic breakthroughs. The open question is whether developers will prioritize population-level fidelity when individual-level coherence already appears sufficient by conventional metrics.

The consequences stratify by identity. Cisgender users face unnecessary pathologization and treatment escalation for transient distress. Transgender users face algorithmic erasure of genuine need, compounding healthcare barriers through safety mechanisms designed to protect them. Clinicians training on LLM-generated cohorts develop diagnostic expectations for patients who exist in no actual population. Until models can faithfully reproduce population distributions rather than individual case plausibility, their deployment for mental health assessment warrants careful consideration of the systematic biases documented here.

13 Statistical Supplement

Dataset Specification:

- Total observations: $N = 28{,}800$
- Unique demographic cohorts: 120
- Iterations per cohort per model: 30
- Models evaluated: 4 (GPT-4o-mini, DeepSeek-V3, Gemini-3-Flash, GLM-4.7)
- Prompt conditions: 2
- Instruments: 4 (PHQ-8, GAD-7, AUDIT-C, PCL-5)

Primary Statistical Tests:

- Stereotype Rigidity: Levene's F-test, all $p < 10^{-40}$
- Threshold Instability: Flip rate 36.66%, drift $t = 15.21$, $p = 7.74 \times 10^{-52}$
- Stochastic Fracture: Kruskal-Wallis $H = 4.89$ (race), $p = 0.295$; $H = 3.43$ (gender), $p = 0.332$ (no significant demographic differences in instability)
- Network Divergence: Maximum $d_F = 0.37$ (Asian), $Z = 6.38$, $p = 8.93 \times 10^{-11}$; Black $d_F = 0.24$, Hispanic $d_F = 0.23$
- Model Family Severity: $t = 10.82$, $p = 3.22 \times 10^{-27}$, $d = 0.13$

Noise Floor Calibration:

- Split-half iterations: 1,000
- Reference group: White ($n = 5{,}760$)
- Mean noise distance: $d_F = 0.17$
- Noise ceiling (95%): $d_F = 0.23$
- Asian and Black exceed ceiling; Hispanic marginally exceeds

Bias Residuals (Cohen's d):

- Trans Woman underestimation: $d = -1.55$ (large)
- Cisgender overestimation: $d = +1.18$ to $+1.21$ (large)
- Racial group overestimation: $d = +1.46$ to $+1.62$ (large)
- SES Low overestimation: $d = +1.91$ (large)
- SES High overestimation: $d = +1.13$ (large)
- Model family severity difference: $d = 0.13$ (small)

Multiple Comparison Correction:

- Method: Benjamini-Hochberg FDR ($\alpha = 0.05$)
- Tests corrected: 47
- Tests surviving correction: 43 (91.5%)

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